# Appendix C

## Item-to-Indicator Mapping for the PLS-SEM Measurement Model

Table C1 reports the complete mapping from questionnaire items, as administered to participants at the school and reproduced in full in Appendix A, to the indicator variables that enter the PLS-SEM measurement model. The second column references the question number in the Appendix A questionnaire; because the ERA-RSI block was administered in three parallel versions depending on the platform selected in question 11 (TikTok: questions 12–37; Instagram: questions 38–63; other platforms: questions 64–88), the three parallel question numbers are reported (TikTok / Instagram / other platforms). The fourth column references the corresponding item number in each source instrument: the Risk of Addiction–Adolescent to Social Networks and Internet Scale (ERA-RSI; Peris et al., 2018), the Pittsburgh Sleep Quality Index (PSQI; Buysse et al., 1989; Spanish adaptations: Moreno et al., 1996; Royuela et al., 2002), the Rosenberg Self-Esteem Scale (RSE; Rosenberg, 1965; Spanish adaptation: Atienza et al., 2000), and the Depression and Anxiety Interference Scale for Children and Adolescents (EIDAN; Espinosa et al., 2022). Items marked (R) in the source-scale column were reverse-coded prior to aggregation. Sociodemographic items (questions 1–2 and 5–10) are not listed, as they do not contribute to any indicator variable but instead function as descriptive or grouping variables in the multigroup analyses (MGA) (Section 4.3); the informed-consent items (questions 3–4) are likewise omitted. Also not listed is the platform-selection item (red_principal, question 11: “Which of these social media platforms do you use the most?”; 0 = I do not use social media, 1 = TikTok, 2 = Instagram, 3 = Facebook, 4 = YouTube, 5 = Other), which served as the questionnaire’s branching variable: participants selecting option 0 were routed to a parallel non-user branch (questions 117–146) and excluded from the analytic sample (see Section 3.3), and the remaining participants

received the ERA-RSI block for their selected platform: questions 12–37 (TikTok), 38–63 (Instagram) or 64–88 (generic social-media wording, for YouTube, Facebook and other services).

**Table C1**

*Item-to-Indicator Mapping From the Questionnaire to the PLS-SEM Measurement Model*

| Indicator | Item (Appendix A) | Content (abbreviated, English) | Source-scale item |
|---|---|---|---|
| era_sint | 13 / 39 / 64 | Daily time spent on [social media platform] (1–4) | ERA-RSI item 1 |
| era_sint | 14 / 40 / 65 | Use [social media platform] during study or work hours | ERA-RSI item 2 |
| era_sint | 15 / 41 / 66 | Would feel angry if forced to give up [social media platform] | ERA-RSI item 3 |
| era_sint | 16 / 42 / 67 | Access [social media platform] anywhere and anytime | ERA-RSI item 4 |
| era_sint | 17 / 43 / 68 | [social media platform] has affected school grades | ERA-RSI item 5 |
| era_sint | 18 / 44 / 69 | Lost sleep due to [social media platform] use | ERA-RSI item 6 |
| era_sint | 19 / 45 / 70 | Hide [social media platform] usage time at home | ERA-RSI item 7 |
| era_sint | 20 / 46 / 71 | Feel insecure without internet access | ERA-RSI item 8 |
| era_sint | 21 / 47 / 72 | Update my status | ERA-RSI item 9 |
| era_uso | 22 / 48 / 73 | Check friends' profiles | ERA-RSI item 10 |
| era_uso | 23 / 49 / 74 | Use the chat function | ERA-RSI item 11 |
| era_uso | 24 / 50 / 75 | Share photos and/or videos | ERA-RSI item 12 |
| era_uso | 25 / 51 / 76 | Comment on photos with friends | ERA-RSI item 13 |
| era_uso | 26 / 52 / 77 | See what contacts have been doing recently | ERA-RSI item 14 |
| era_uso | 27 / 53 / 78 | Reconnect with former friends | ERA-RSI item 15 |
| era_uso | 28 / 54 / 79 | Number of videos published (rescaled 0–3 → 1–4) | ERA-RSI item 16 |
| era_uso | 29 / 55 / 80 | Make new friends | ERA-RSI item 17 |
| era_fri | 30 / 56 / 81 | Play virtual and/or role-playing games | ERA-RSI item 18 |
| era_fri | 31 / 57 / 82 | Join special-interest groups | ERA-RSI item 19 |
| era_nomo | 32 / 58 / 83 | Feel safer being able to communicate anytime | ERA-RSI item 24 |
| era_nomo | 33 / 59 / 84 | Safer to send a photo by phone than on social media | ERA-RSI item 25 |
| era_nomo | 34 / 60 / 85 | Anxiety when messages are not answered immediately | ERA-RSI item 26 |
| era_nomo | 35 / 61 / 86 | Would feel furious if mobile phone were taken away | ERA-RSI item 27 |
| era_nomo | 36 / 62 / 87 | Need to know whether the recipient has read my message | ERA-RSI item 28 |
| era_nomo | 37 / 63 / 88 | Uneasy when no one talks to me online | ERA-RSI item 29 |
| psqi | 89 | Usual bedtime (last month) | PSQI item p1 (C4) |
| psqi | 90 | Time to fall asleep, in minutes (last month) | PSQI item p2 (C2) |
| psqi | 91 | Usual wake-up time (last month) | PSQI item p3 (C4) |
| psqi | 92 | Actual hours of sleep per night (last month) | PSQI item p4 (C3, C4) |
| psqi | 93 | Trouble sleeping: cannot fall asleep within 30 min | PSQI item p5a (C2) |
| psqi | 94 | Trouble sleeping: wake during the night or early morning | PSQI item p5b (C5) |

| Indicator | Item (Appendix A) | Content (abbreviated, English) | Source-scale item |
|---|---|---|---|
| psqi | 95 | Trouble sleeping: need to get up to use the bathroom | PSQI item p5c (C5) |
| psqi | 96 | Trouble sleeping: cannot breathe comfortably | PSQI item p5d (C5) |
| psqi | 97 | Trouble sleeping: cough or snore loudly | PSQI item p5e (C5) |
| psqi | 98 | Trouble sleeping: feel too cold | PSQI item p5f (C5) |
| psqi | 99 | Trouble sleeping: feel too hot | PSQI item p5g (C5) |
| psqi | 100 | Trouble sleeping: had bad dreams or nightmares | PSQI item p5h (C5) |
| psqi | 101 | Trouble sleeping: had pain | PSQI item p5i (C5) |
| psqi | 102 | Trouble sleeping: other reasons (free text; excluded from C5) | PSQI item p5j (excluded) |
| psqi | 103 | Overall subjective sleep quality (last month) | PSQI item p6 (C1) |
| psqi | 104 | Use of sleep medication (last month) | PSQI item p7 (C6) |
| psqi | 105 | Daytime sleepiness at school or other activities | PSQI item p8 (C7) |
| psqi | 106 | Lack of motivation for daily activities | PSQI item p9 (C7) |
| Self-Esteem | 107 | I feel I am a person of worth, equal to others | RSE item 1 |
| Self-Esteem | 108 | I tend to feel that I am a failure (reverse-coded) | RSE item 2 (R) |
| Self-Esteem | 109 | I feel I have a number of good qualities | RSE item 3 |
| Self-Esteem | 110 | I am able to do things as well as most other people | RSE item 4 |
| Self-Esteem | 111 | I feel I do not have much to be proud of (reverse-coded) | RSE item 5 (R) |
| Self-Esteem | 112 | I take a positive attitude toward myself | RSE item 6 |
| Self-Esteem | 113 | On the whole, I am satisfied with myself | RSE item 7 |
| Self-Esteem | 114 | I wish I could have more respect for myself (reverse-coded) | RSE item 8 (R) |
| Self-Esteem | 115 | I certainly feel useless at times (reverse-coded) | RSE item 9 (R) |
| Self-Esteem | 116 | At times I think I am no good at all (reverse-coded) | RSE item 10 (R) |
| eidan_school | 148 | Difficulty doing homework or studying | EIDAN item 2 (F1) |
| eidan_school | 150 | Difficulty with daily activities (eating, sleeping, etc.) | EIDAN item 4 (F1) |
| eidan_school | 152 | Did not perform on exams as well as could have | EIDAN item 6 (F1) |
| eidan_school | 155 | Difficulty paying attention or asking questions in class | EIDAN item 9 (F1) |
| eidan_family | 147 | Difficulty getting along with siblings or other family | EIDAN item 1 (F2) |
| eidan_family | 151 | Difficulty communicating with parents | EIDAN item 5 (F2) |
| eidan_family | 154 | Difficulty doing pleasant activities with family | EIDAN item 8 (F2) |
| eidan_peers | 149 | Difficulty relating to classmates at school | EIDAN item 3 (F3) |
| eidan_peers | 153 | Difficulty or reluctance to meet up with friends | EIDAN item 7 (F3) |
| eidan_peers | 156 | Difficulty making new friends | EIDAN item 10 (F3) |

*Note.* Indicator names match the variable labels used in the PLS-SEM specification (see Table 1). Question numbers correspond to the order of presentation in the institutional questionnaire reproduced in Appendix A; for ERA-RSI items, the three numbers refer to the TikTok, Instagram, and other-platforms branches, respectively. Source-scale item numbers refer to the canonical numbering in the respective validation studies. ERA-RSI

= Risk of Addiction–Adolescent to Social Networks and Internet Scale (Peris et al., 2018); PSQI = Pittsburgh Sleep Quality Index (Buysse et al., 1989); RSE = Rosenberg Self-Esteem Scale (Rosenberg, 1965); EIDAN = Depression and Anxiety Interference Scale for Children and Adolescents (Espinosa et al., 2022). (R) = reverse-coded prior to aggregation. F1, F2, and F3 denote the three factors of the EIDAN scale (school and daily activities, family, peers, respectively). C1–C7 denote the seven PSQI components (subjective sleep quality, sleep latency, sleep duration, habitual sleep efficiency, sleep disturbances, use of sleep medication, and daytime dysfunction), derived from the listed items following the scoring algorithm of Buysse et al. (1989) and summed to yield the global score. [social media platform] denotes the participant's primary platform, as selected via the branching item red_principal, and item wording was tailored accordingly. Items 20–23 of the original ERA-RSI freaky-traits subscale (erotic-sexual content) were excluded a priori from the questionnaire on ethical grounds. The device item (questions 12 and 38; not administered in the other-platforms branch), which records the device used to access the social media platform, is contextual and does not enter the computation of any indicator variable; it is therefore not listed. Question 102 (PSQI p5j) was administered as free text and is therefore not included in the computation of PSQI component C5.

**References**

Atienza, F. L., Moreno, Y., & Balaguer, I. (2000). An analysis of the dimensionality of the Rosenberg self-esteem scale in a sample of Valencian adolescents. *Revista de Psicología*, *22*(12), 29-42.

Buysse, D. J., Reynolds III, C. F., Monk, T. H., Berman, S. R., & Kupfer, D. J. (1989). The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. Psychiatry research, 28(2), 193-213.

Espinosa, V., Sandin, B., Garcia-Escalera, J., Valiente, R. M., & Chorot, P. (2022). Depression and Anxiety Interference Scale for Children and Adolescents: Factorial Structure and Psychometric Properties/Escala de Interferencia de la Depresión y la Ansiedad para Niños y Adolescentes (EIDAN): Estructura Factorial y Propiedades Psicométricas. *Revista Iberoamericana de Diagnóstico y Evaluación-e Avaliação Psicológica*, (65), 19-34.

Moreno, P. L., Aguado, J. A., Royuela, A., & Macías, J. A. (1996). Influence of examinations in sleep quality of the students in medicine in Valladolid evaluated by PSQI. *European Psychiatry*, *11*(S4), 392s-392s.

Peris, M., Maganto, C., & Garaigordobil, M. (2018). Scale of risk of addiction to social networks and Internet for adolescents: reliability and validity (ERA-RSI). Revista de Psicología Clínica con Niños y Adolescentes, 5(2), 30-36.

Rosenberg, M. (1965). *Society and the adolescent self-image* (Vol. 11, p. 326). Princeton, NJ: Princeton university press.

Royuela, A., Macias, J. A., Gil-Verona, J. A., Pastor, J. F., Maniega, M. A., Alonso, J., ... & Boget, T. (2002). Sleep in schizophrenia: a preliminary study using the Pittsburgh Sleep Quality Index. *Neurobiology of Sleep-Wakefulness Cycle*, *2*(2), 37-39.